

human

Show me natural areas in Exeter?

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to see natural areas in Exeter. Provide the general steps to solve this query.'}

planning_agent

The area specified is Exeter, so there is 1 entry for Exeter. The entity to search for is natural areas, and we need to find as many natural areas as possible within Exeter.

output steps: ["Find Exeter 1 area", "Search for natural areas in Exeter as many search results"]

host_agent

Tool send_message returned result : The area specified is Exeter, so there is 1 entry for Exeter. The entity to search for is natural areas, and we need to find as many natural areas as possible within Exeter.

output steps: ["Find Exeter 1 area", "Search for natural areas in Exeter as many search results"]

Calling tool generate_metadata_for_all_artifacts with args : {} Tool generate_metadata_for_all_artifacts returned result : {} Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']} Tool generate_metadata_for_artifacts returned result : [...]

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using all relevant columns
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower() == 'exeter city') |
            (gdf['name1_text'].str.lower() == 'exeter')
        ) &
        (
            (gdf['descriptiongroup'].str.lower().isin(['settlement', 'part of settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town', 'suburban area']))
        )
    ]
    summary = f"Filtered Exeter area polygon: {len(filtered)} entries found."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) for Exeter area, filtered by name and settlement/city/town/suburban area type.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered Exeter area polygon: 1 entries found.',]

Output from agent data_analysis_agent : Filtered Exeter area polygon: 1 entry found. An artifact named 'filtered_exeter_area_polygon' has been generated, containing only the polygon(s) for the Exeter area, filtered by name and settlement/city/town/suburban area type.

Tool send_message returned result : ["Filtered Exeter area polygon: 1 entry found.\nAn artifact named 'filtered_exeter_area_polygon' has been generated, containing only the polygon(s) for the Exeter area, filtered by name and settlement/city/town/suburban area type.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) for Exeter area, filtered by name and settlement/city/town/suburban area type.']", []]

named_area_agent

Output from agent named_area : The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This artifact contains only the polygon(s) for the Exeter area, filtered by name and settlement/city/town/suburban area type.

Tool send_message returned result : ["The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This artifact contains only the polygon(s) for the Exeter area, filtered by name and settlement/city/town/suburban area type.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) for Exeter area, filtered by name and settlement/city/town/suburban area type.']", []]

host_agent

Calling tool send_message with args : {'target': 'land_features', 'task_description': "Search for natural land features (such as coniferous trees, cliffs, heath, rough grassland, etc.) within the 'filtered_exeter_area_polygon' artifact."}

land_features_agent

Calling tool call_os_ngd with args : {'filters': ['Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'natural_land_features_exeter'} Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['\land_natural_land_features_exeter\', '\landform_natural_land_features_exeter\'],\n Descriptions are : ["A geopandas dataframe containing land data with filters and bbox applied for the query Search for natural land features (such as coniferous trees, cliffs, heath, rough grassland, etc.) within the 'filtered_exeter_area_polygon' artifact. using filters ['Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland']. ", "A geopandas dataframe containing land form data with filters and bbox applied for the query Search for natural land features (such as coniferous trees, cliffs, heath, rough grassland, etc.) within the 'filtered_exeter_area_polygon' artifact. using filters ['Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland']."],\n counts of records fetched are : [18, 19].\n ', [,]]

Output from agent land_features : I found two datasets related to natural land features within the 'filtered_exeter_area_polygon':

land_natural_land_features_exeter: Contains land data with filters applied for coniferous trees, cliffs, heath, and rough grassland. (18 records)

landform_natural_land_features_exeter: Contains landform data with the same filters. (19 records)

Would you like further analysis on these datasets?

Tool send_message returned result : ['I found two datasets related to natural land features within the 'filtered_exeter_area_polygon':\n\n1. **land_natural_land_features_exeter**: Contains land data with filters applied for coniferous trees, cliffs, heath, and rough grassland. (18 records)\n\n2. **landform_natural_land_features_exeter**: Contains landform data with the same filters. (19 records)\n\nWould you like further analysis on these datasets?Additionally some data artifacts have been generated with names ['land_natural_land_features_exeter', 'landform_natural_land_features_exeter'] and \n descriptions ["A geopandas dataframe containing land data with filters and bbox applied for the query Search for natural land features (such as coniferous trees, cliffs, heath, rough grassland, etc.) within the 'filtered_exeter_area_polygon' artifact. using filters ['Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland']. ", "A geopandas dataframe containing land form data with filters and bbox applied for the query Search for natural land features (such as coniferous trees, cliffs, heath, rough grassland, etc.) within the 'filtered_exeter_area_polygon' artifact. using filters ['Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland']."]', [,]]

host_agent

Calling tool send_message with args : {'target': 'land_use_features', 'task_description': "Search for land use features that represent natural areas (such as parks, nature reserves, etc.) within the 'filtered_exeter_area_polygon' artifact."}

land_use_features_agent

Calling tool call_os_ngd with args : {'filters': ['Amenity And Open Space Site', 'Arboretum', 'Botanical Garden', 'Nature Reserve', 'Park And Ride Car Park', 'Public Park Or Garden', 'Recreation Ground', 'Wildlife Observation Site', 'Wildlife Or Zoological Park'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'natural_areas_exeter'} Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['landuse_natural_areas_exeter'],\n Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for land use features that represent natural areas (such as parks, nature reserves, etc.) within the 'filtered_exeter_area_polygon' artifact. using filters ['Amenity And Open Space Site', 'Arboretum', 'Botanical Garden', 'Nature Reserve', 'Park And Ride Car Park', 'Public Park Or Garden', 'Recreation Ground', 'Wildlife Observation Site', 'Wildlife Or Zoological Park']. Further Name filtering is available for this"],\n counts of records fetched are : [44].\n ', []]

Output from agent land_use_features : I found 44 records of land use features representing natural areas within the 'filtered_exeter_area_polygon'. The data is saved in the artifact named 'landuse_natural_areas_exeter'.

Would you like any further analysis on this data?

Tool send_message returned result : ['I found 44 records of land use features representing natural areas within the \'filtered_exeter_area_polygon\'. The data is saved in the artifact named \'landuse_natural_areas_exeter\'.\n\nWould you like any further analysis on this data?Additionally some data artifacts have been generated with names [\'landuse_natural_areas_exeter\'] and \n descriptions ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for land use features that represent natural areas (such as parks, nature reserves, etc.) within the \'filtered_exeter_area_polygon\' artifact. using filters [\'Amenity And Open Space Site\', \'Arboretum\', \'Botanical Garden\', \'Nature Reserve\', \'Park And Ride Car Park\', \'Public Park Or Garden\', \'Recreation Ground\', \'Wildlife Observation Site\', \'Wildlife Or Zoological Park\']. Further Name filtering is available for this"]', []]

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the following artifacts to show natural areas in Exeter: 'land_natural_land_features_exeter', 'landform_natural_land_features_exeter', and 'landuse_natural_areas_exeter'. Overlay these on the 'filtered_exeter_area_polygon' to show their location within Exeter."}

plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['land_natural_land_features_exeter', 'landform_natural_land_features_exeter', 'landuse_natural_areas_exeter', 'filtered_exeter_area_polygon']} Tool generate_metadata_for_artifacts returned result : [...]

Tool Code Executor

```
def plot_natural_areas_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack the data
    land_natural_land_features = data[0]
    landform_natural_land_features = data[1]
    landuse_natural_areas = data[2]
    exeter_polygon = data[3]

    # Ensure all are in EPSG:4326
    land_natural_land_features = land_natural_land_features.to_crs(epsg=4326)
    landform_natural_land_features = landform_natural_land_features.to_crs(epsg=4326)
    landuse_natural_areas = landuse_natural_areas.to_crs(epsg=4326)
    exeter_polygon = exeter_polygon.to_crs(epsg=4326)

    # Clip all features to Exeter polygon
```

```

exeter_geom = exeter_polygon.iloc[0].geometry
land_natural_land_features_clip = gpd.clip(land_natural_land_features, exeter_geom)
landform_natural_land_features_clip = gpd.clip(landform_natural_land_features, exeter_geom)
landuse_natural_areas_clip = gpd.clip(landuse_natural_areas, exeter_geom)

# Get centroid for map
centroid = exeter_geom.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=12, tiles='cartodbpositron')

# Add Exeter boundary
folium.GeoJson(
    mapping(exeter_geom),
    name='Exeter Boundary',
    style_function=lambda x: {'color': 'black', 'weight': 3, 'fillOpacity': 0}
).add_to(m)

# Add land_natural_land_features
folium.GeoJson(
    land_natural_land_features_clip,
    name='Natural Land Features',
    style_function=lambda x: {'color': 'green', 'weight': 2, 'fillOpacity': 0.3}
).add_to(m)

# Add landform_natural_land_features
folium.GeoJson(
    landform_natural_land_features_clip,
    name='Landform Natural Features',
    style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillOpacity': 0.3}
).add_to(m)

# Add landuse_natural_areas
folium.GeoJson(
    landuse_natural_areas_clip,
    name='Natural Areas (Land Use)',
    style_function=lambda x: {'color': 'orange', 'weight': 2, 'fillOpacity': 0.3}
).add_to(m)

folium.LayerControl().add_to(m)

# Save map
map_filename = 'natural_areas_exeter_map.html'
m.save(map_filename)

# Prepare summary
summary = (
    f"Map of natural areas in Exeter.\n"
    f"- Land Natural Land Features: {len(land_natural_land_features_clip)} polygons\n"
    f"- Landform Natural Land Features: {len(landform_natural_land_features_clip)} polygons\n"

```



```

f"- Landuse Natural Areas: {len(landuse_natural_areas_clip)} polygons\n"
f"First 5 Land Natural Land Features: {land_natural_land_features_clip['description'].head(5).tolist()}\n"
f"First 5 Landform Natural Land Features: {landform_natural_land_features_clip['description'].head(5).tolist()}\n"
f"First 5 Landuse Natural Areas: {landuse_natural_areas_clip['description'].head(5).tolist()}"
)

return [
    summary,
    'natural_areas_exeter_map',
    'Folium map showing natural land features, landform features, and natural land use areas within the Exeter city boundary.',
    map_filename
]

```

Tool code_executor returned result : Traceback (most recent call last): File "c:\Users\lab1574\OneDrive - University of Exeter\Desktop\Ordnance_Survey\utils\tools.py", line 253, in code_executor output = namespacefunction_name File "", line 38, in plot_natural_areas_exeter File "c:\Users\lab1574\OneDrive - University of Exeter\Desktop\Ordnance_Survey\osvenv\Lib\site-packages\folium\features.py", line 733, in init self.data = self.process_data(data) File "c:\Users\lab1574\OneDrive - University of Exeter\Desktop\Ordnance_Survey\osvenv\Lib\site-packages\folium\features.py", line 776, in process_data return json.loads(json.dumps(data.geo_interface)) File "C:\Users\lab1574\AppData\Local\Programs\Python\Python311\Lib\json__init__.py", line 231, in dumps return _default_encoder.encode(obj) File "C:\Users\lab1574\AppData\Local\Programs\Python\Python311\Lib\json\encoder.py", line 200, in encode chunks = self.iterencode(o, _one_shot=True) File "C:\Users\lab1574\AppData\Local\Programs\Python\Python311\Lib\json\encoder.py", line 258, in iterencode return _iterencode(o, 0) File "C:\Users\lab1574\AppData\Local\Programs\Python\Python311\Lib\json\encoder.py", line 180, in default raise TypeError(f'Object of type {o.class.name} ' TypeError: Object of type Timestamp is not JSON serializable

Tool Code Executor

```

def plot_natural_areas_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack the data
    land_natural_land_features = data[0]
    landform_natural_land_features = data[1]
    landuse_natural_areas = data[2]
    exeter_polygon = data[3]

```

```

# Ensure all are in EPSG:4326
land_natural_land_features = land_natural_land_features.to_crs(epsg=4326)
landform_natural_land_features = landform_natural_land_features.to_crs(epsg=4326)
landuse_natural_areas = landuse_natural_areas.to_crs(epsg=4326)
exeter_polygon = exeter_polygon.to_crs(epsg=4326)

# Clip all features to Exeter polygon
exeter_geom = exeter_polygon.iloc[0].geometry
land_natural_land_features_clip = gpd.clip(land_natural_land_features, exeter_geom)
landform_natural_land_features_clip = gpd.clip(landform_natural_land_features, exeter_geom)
landuse_natural_areas_clip = gpd.clip(landuse_natural_areas, exeter_geom)

# Remove problematic columns for folium (non-serializable)
def drop_nonserializable(gdf):
    for col in gdf.columns:
        if gdf[col].dtype.name.startswith('datetime'):
            gdf = gdf.drop(columns=[col])
    return gdf
land_natural_land_features_clip = drop_nonserializable(land_natural_land_features_clip)
landform_natural_land_features_clip = drop_nonserializable(landform_natural_land_features_clip)
landuse_natural_areas_clip = drop_nonserializable(landuse_natural_areas_clip)

# Get centroid for map
centroid = exeter_geom.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=12, tiles='cartodbpositron')

# Add Exeter boundary
folium.GeoJson(
    mapping(exeter_geom),
    name='Exeter Boundary',
    style_function=lambda x: {'color': 'black', 'weight': 3, 'fillOpacity': 0}
).add_to(m)

# Add land_natural_land_features
folium.GeoJson(
    land_natural_land_features_clip,
    name='Natural Land Features',
    style_function=lambda x: {'color': 'green', 'weight': 2, 'fillOpacity': 0.3}
).add_to(m)

# Add landform_natural_land_features
folium.GeoJson(
    landform_natural_land_features_clip,
    name='Landform Natural Features',
    style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillOpacity': 0.3}
).add_to(m)

# Add landuse_natural_areas

```



```

folium.GeoJson(
    landuse_natural_areas_clip,
    name='Natural Areas (Land Use)',
    style_function=lambda x: {'color': 'orange', 'weight': 2, 'fillOpacity': 0.3}
).add_to(m)

folium.LayerControl().add_to(m)

# Save map
map_filename = 'natural_areas_exeter_map.html'
m.save(map_filename)

# Prepare summary
summary = (
    f"Map of natural areas in Exeter.\n"
    f"- Land Natural Land Features: {len(land_natural_land_features_clip)} polygons\n"
    f"- Landform Natural Land Features: {len(landform_natural_land_features_clip)} polygons\n"
    f"- Landuse Natural Areas: {len(landuse_natural_areas_clip)} polygons\n"
    f"First 5 Land Natural Land Features: {land_natural_land_features_clip['description'].head(5).tolist()}\n"
    f"First 5 Landform Natural Land Features: {landform_natural_land_features_clip['description'].head(5).tolist()}\n"
    f"First 5 Landuse Natural Areas: {landuse_natural_areas_clip['description'].head(5).tolist()}"
)

return [
    summary,
    'natural_areas_exeter_map',
    'Folium map showing natural land features, landform features, and natural land use areas within the Exeter city boundary.',
    map_filename
]

```

Tool code_executor returned result : ["Map of natural areas in Exeter.\n- Land Natural Land Features: 18 polygons\n- Landform Natural Land Features: 19 polygons\n- Landuse Natural Areas: 44 polygons\nFirst 5 Land Natural Land Features: ['Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees']\nFirst 5 Landform Natural Land Features: ['Cliff', 'Cliff', 'Cliff', 'Cliff', 'Cliff']\nFirst 5 Landuse Natural Areas: ['Amenity And Open Space Site', 'Park And Ride Car Park', 'Amenity And Open Space Site', 'Public Park Or Garden', 'Recreation Ground']",]

Output from agent plotting_agent : Map of natural areas in Exeter. - Land Natural Land Features: 18 polygons - Landform Natural Land Features: 19 polygons - Landuse Natural Areas: 44 polygons First 5 Land Natural Land Features: ['Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees'] First 5 Landform Natural Land Features: ['Cliff', 'Cliff', 'Cliff', 'Cliff', 'Cliff'] First 5 Landuse Natural Areas: ['Amenity And Open Space Site', 'Park And Ride Car Park', 'Amenity And Open Space Site', 'Public Park Or Garden', 'Recreation Ground']

Artifact name: natural_areas_exeter_map

Description: Folium map showing natural land features, landform features, and natural land use areas within the Exeter city boundary.

Map file: natural_areas_exeter_map.html

Tool send_message returned result : ["Map of natural areas in Exeter.\n- Land Natural Land Features: 18 polygons\n- Landform Natural Land Features: 19 polygons\n- Landuse Natural Areas: 44 polygons\nFirst 5 Land Natural Land Features: ['Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees', 'Coniferous Trees']\nFirst 5 Landform Natural Land Features: ['Cliff', 'Cliff', 'Cliff', 'Cliff', 'Cliff']\nFirst 5 Landuse Natural Areas: ['Amenity And Open Space Site', 'Park And Ride Car Park', 'Amenity And Open Space Site', 'Public Park Or Garden', 'Recreation Ground']\n\nArtifact name: natural_areas_exeter_map\n\nDescription: Folium map showing natural land features, landform features, and natural land use areas within the Exeter city boundary.\n\nMap file: natural_areas_exeter_map.html\n\nAdditionally some data artifacts have been generated with names ['natural_areas_exeter_map'] and\n\n descriptions ['Folium map showing natural land features, landform features, and natural land use areas within the Exeter city boundary.']", []]

host_agent

Output from agent host_agent : Here are the natural areas in Exeter:

- Natural land features (such as coniferous trees, cliffs, heath, and rough grassland)
- Landform features (such as cliffs and other natural landforms)
- Land use features representing natural areas (such as parks, nature reserves, public gardens, and open spaces)

All these features have been mapped within the Exeter city boundary. The map includes: - 18 polygons of natural land features - 19 polygons of natural landform features - 44 polygons of natural land use areas

If you would like to see the map or need further details about any specific type of natural area, let me know!